Filecoin Network Supply Forecast: Scenario Analysis

CEL — February 2026

Executive Summary

This document presents a two-year forward simulation of the Filecoin network under two scenarios that reflect the current trajectory of declining storage provider (SP) onboarding.

The central finding is that **locked FIL declines under both scenarios**, with the severity depending on whether renewal rates hold steady or deteriorate. Circulating supply approaches 1 billion FIL in both cases. **Based on the trajectory of the past year, the Decline scenario is the more realistic projection.**

Table 1: Forecast summary. "Current" reflects on-chain values as of February 22, 2026. Status Quo and Decline columns show values at the end of the two-year forecast horizon.

Metric	Current	Status Quo	Decline
Network Locked (M FIL)	101.7	71.6	60.0
Circulating Supply (M FIL)	836.9	951.2	962.1
Locked / Circ Supply	0.122	0.075	0.062
Pledge per 32 GiB Sector (FIL)	0.172	0.282	0.429
1Y Forward FoFR (%)	23.1	28.9	54.8

Implications and Recommendations

1. **Circulating supply is approaching 1 billion FIL.** Under both scenarios, circulating supply crosses 900M FIL within the forecast horizon and approaches 960M under Decline. With locked FIL contracting and minting ongoing, the network is on track to cross the 1B threshold. Once crossed, there is no protocol mechanism to reverse this — the only path back is increased locking demand.

2. **The network needs an alternative locking mechanism urgently.** Sector pledge is the only meaningful source of FIL locking today, and it is shrinking. Implementing **veFIL** (or an

equivalent vote-escrowed locking mechanism) would create a second source of locking demand that is not tied to storage onboarding. This should be treated as a high-priority initiative.

3. **It is worse for Filecoin if no one wants to onboard capacity than if there are no paid deals.** The onboarding decline is the more existential threat. Paid storage deals are desirable, but the network's economic security depends on SPs continuing to onboard and renew sectors regardless of deal status. Policy decisions should prioritize removing barriers to onboarding above all else.

4. **FIL+ should be reconsidered.** Filecoin no longer provides an effective subsidy to SPs — even with paid deals, the economics are marginal. FIL+ has become an administrative burden across the ecosystem (notaries, allocators, compliance) without delivering commensurate onboarding growth. We should strongly consider removing FIL+ entirely by granting all sectors the 10x quality multiplier by default. This would eliminate friction for SPs, reduce governance overhead, and acknowledge the reality that the verified-deal distinction no longer drives meaningful network growth.

Scenario Definitions

The simulation is initialized with 365 days of on-chain history (February 2025 – February 2026). Key inputs are derived from 30-day trailing statistics.

Current network parameters (30-day):

- Raw byte onboarding: 1.01 PiB/day (median)
- Renewal rate: 72.8% (mean)
- FIL+ rate: 97.1% (median)

Both scenarios share the same onboarding trajectory: a 3x per year exponential decay, which matches the observed decline over the past 12 months. Both scenarios also assume declining renewal rates, reflecting the expectation that rising pledge requirements will deter marginal SPs from renewing. The scenarios differ in the severity of this renewal rate erosion.

Both renewal rate curves follow a front-loaded (concave) shape: the decline is steepest in the early period, reflecting the expectation that marginal SPs — those most sensitive to rising costs — exit first, with the rate of decline flattening as only committed operators remain.

Status Quo — Onboarding continues its observed decline (3x/yr). The renewal rate experiences mild erosion, declining from 73% to 55% over two years. This represents the case where current conditions persist: no new growth materializes, and rising pledge requirements gradually push out the least committed SPs.

Decline — Same onboarding trajectory, but renewal rate erosion is more severe, declining from 73% to 36% over two years. This scenario reflects accelerating SP attrition as the negative feedback loop between network contraction and rising pledge requirements intensifies.

Table 2: Scenario input parameters. Both scenarios share declining onboarding and front-loaded renewal rate curves.

Parameter	Status Quo	Decline
RB Onboarding	1.01 $\rightarrow$ 0.34 (1Y) $\rightarrow$ 0.11 (2Y) PiB/d	Same
Renewal Rate	73% $\rightarrow$ 60% (1Y) $\rightarrow$ 55% (2Y)	73% $\rightarrow$ 47% (1Y) $\rightarrow$ 36% (2Y)
FIL+ Rate	97.1% (flat)	97.1% (flat)

Forecast Results

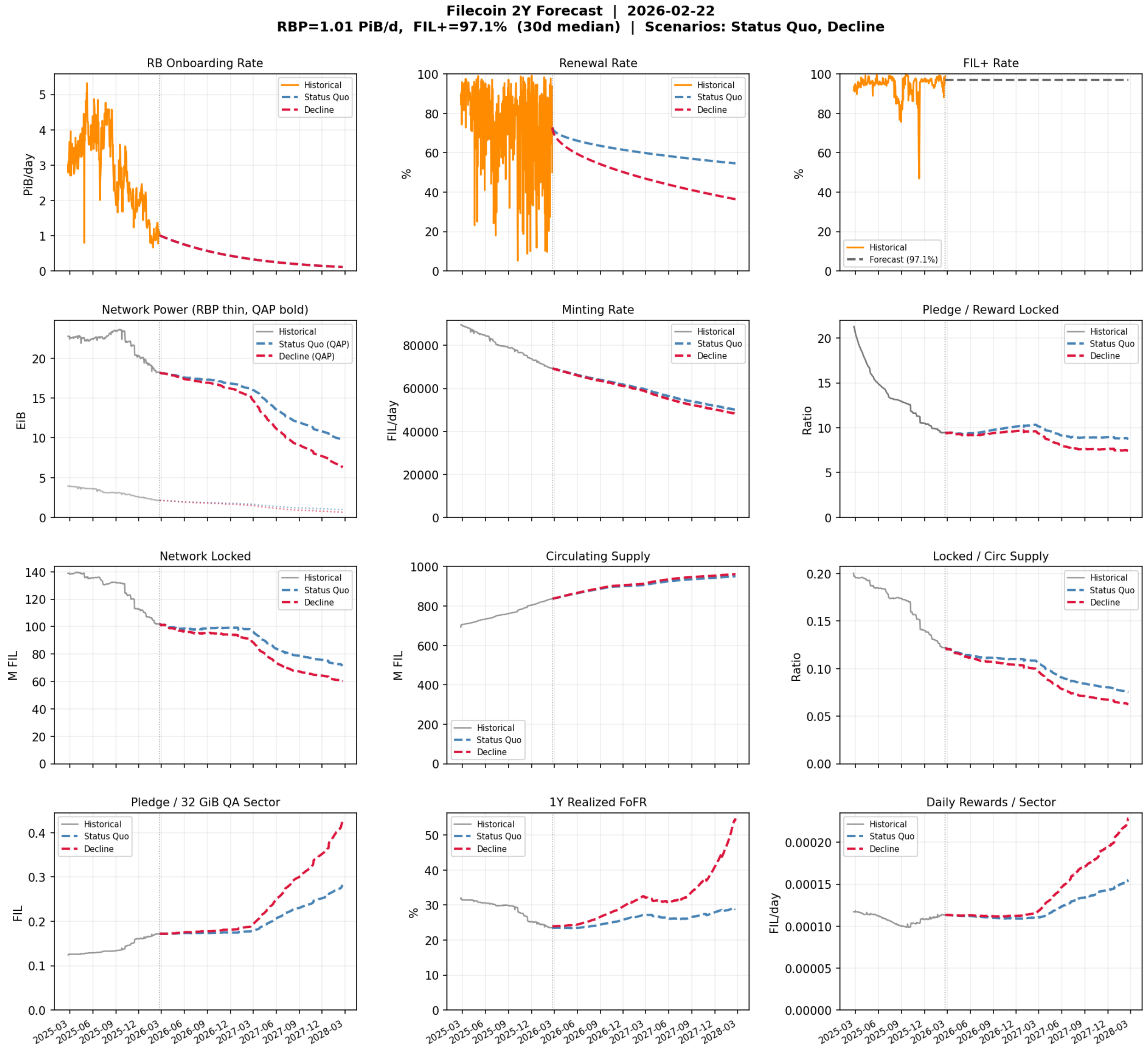

Figure 1: Two-year scenario forecast panel. Rows from top: inputs, network power and minting, supply metrics, and SP economics. The vertical dotted line marks the transition from historical data (solid gray) to the forecast period (dashed, color-coded by scenario).

Network Power

Under the Status Quo scenario, quality-adjusted power (QAP) declines from 18.2 EiB to approximately 11.8 EiB (-35%). Even with only mild renewal rate erosion, the falling rate of new onboarding means the network cannot replace expiring capacity fast enough to maintain its current size.

The Decline scenario shows QAP falling to approximately 6.4 EiB (-65%) as the front-loaded decline in renewal rate compounds the onboarding shortfall. Sectors that would have renewed under flat conditions increasingly exit as rising pledge requirements make renewal uneconomic.

Locked FIL and Circulating Supply

Locked FIL is the most consequential metric for circulating supply dynamics.

Under the **Status Quo**, locked FIL declines from 101.7M to 71.6M (-30%). Circulating supply grows from 837M to 951M (+14%), driven by ongoing minting and vesting. L/CS declines from 0.122 to 0.075 (-38%).

Under the **Decline** scenario, the trajectory is more severe:

Table 3: Decline scenario — cumulative change from forecast start.

Horizon	QAP	Locked FIL	Circ Supply	L/CS
Start	18.2 EiB	101.7M	836.9M	0.122
+6 months	-7%	-6%	+6%	-12%
+1 year	-17%	-11%	+9%	-19%
+18 months	-49%	-34%	+13%	-41%
+2 years	-65%	-41%	+15%	-49%

- **First year (decline begins immediately):** The front-loaded drop in renewal rate means locked FIL begins declining from the start, reaching -11% after one year. The 360-day sector duration still creates some inertia — QAP contracts 17% while locked FIL trails at -11% — but the steeper early RR decline ensures there is no false plateau. L/CS declines 19%.
- **Second year (acceleration phase):** Sectors locked during the transition begin to expire. With the renewal rate now approaching 36%, roughly two-thirds of expiring sectors release their pledge. Locked FIL drops 41% while circulating supply continues to grow (+15%). L/CS falls 49%, compounded by both the declining numerator and growing denominator.

SP Economics

Pledge per sector exhibits an inverse relationship with network size under attrition. As QAP contracts, the consensus pledge component — which scales with the ratio of sector size to total network power — increases. Under Status Quo, pledge rises from 0.17 to 0.28 FIL per 32 GiB sector (+64%); under Decline, it reaches 0.43 FIL (+150%). This creates a negative feedback loop: rising pledge deters renewals, which shrinks the network, which raises pledge further.

The 1-year forward rate of return (FoFR) rises to approximately 29% under Status Quo and 55% under Decline, up from 23%. This is mechanical: fewer sectors compete for block rewards that decline more slowly than network power. This elevated return represents the equilibrium incentive for remaining providers but may be insufficient to attract new entrants if accompanied by adverse token price dynamics.

Key Takeaways

1. **Locked FIL declines under both scenarios.** Under Status Quo with mild renewal rate erosion, locked FIL falls 30% over two years. Under accelerating attrition, the decline reaches 41%. There is no stability scenario — both paths lead to contraction.

2. **The Decline scenario is the more realistic projection.** Raw byte onboarding has already declined ~3x over the past year, and rising pledge requirements create a negative feedback loop that is likely to erode renewal rates. The Decline scenario projects L/CS compressing from 0.122 to 0.062 over two years.

3. **Locked FIL has significant inertia.** The 360-day sector duration creates a lag between declining conditions and their effect on locked FIL. Even in the Decline scenario, locked FIL is down only 11% after one year despite QAP already contracting 17%. The full impact materializes in the second year as sectors locked during the transition begin to expire without renewal.

4. **Both onboarding and renewal rate drive the decline, but renewals dominate the flow.** The daily volume of power flowing through renewals (expiring sectors being re-locked) is roughly 10x larger than new onboarding in power terms. Changes in renewal behavior therefore have an outsized effect on locked FIL relative to changes in new onboarding.

5. **Attrition is self-reinforcing in pledge terms.** As the network shrinks, pledge per sector rises (+64% under Status Quo, +150% under Decline), which may further discourage participation. FoFR also rises as a partial offset, but the net effect on SP decision-making depends on token price dynamics not modeled here.

6. **Circulating supply is bounded but approaching 1B FIL.** Under both scenarios, circulating supply grows to 950–962M FIL over two years. Both scenarios approach the 1 billion FIL threshold, a level with no protocol mechanism for reversal.

7. **These projections are likely still optimistic.** The model's renewal mechanism recalculates pledge at current conditions and retains the higher of original and recalculated pledge. In practice, SPs facing substantially higher pledge requirements may choose not to renew — a behavioral response the model does not capture. The on-chain locked FIL decline rate is currently steeper than even the Decline scenario projects for its first 90 days, suggesting the true trajectory may be worse than modeled here.

Appendix: Model Notes

- **Data source:** Spacescope API (on-chain supply and power statistics)
- **History window:** 365 days (Feb 2025 – Feb 2026)
- **Forecast horizon:** 730 days (Feb 2026 – Feb 2028)
- **Onboarding trajectory:** 3x/yr exponential decay, matching the observed historical decline in raw byte onboarding over the past 12 months.

- **Renewal rate:** 30-day mean (72.8%). The mean is used rather than the median (82.1%) because locked FIL is a cumulative stock: total pledge retained = $\sum_i \mathrm{RR}_i \times \mathrm{scheduled\_expiration}_i \approx \overline{\mathrm{RR}} \times \sum_i \mathrm{scheduled\_expiration}_i$. The mean captures the full distribution including low-RR days that the median would discard.
- **FIL+ rate:** Held constant at 97.1% across all scenarios.
- **Sector duration:** 360 days (average).
- **Initialization:** Pledge/reward split estimated from 180-day linear vesting integral of on-chain minting data (4.5% reward-locked, 95.5% pledge-locked).
- **Renewal rate curve shape:** Both scenarios use front-loaded (concave) renewal rate curves rather than linear ramps. This reflects the expectation that marginal SPs — those most sensitive to rising costs — exit first, producing a steeper initial decline that flattens as only committed operators remain.
- **Why declining renewal rates rather than flat?** Previous forecasts held renewal rate constant at the observed 30-day mean or median. This was appropriate when the network was growing or stable, but Filecoin is now in a fundamentally different regime: onboarding is in sustained decline, pledge per sector is rising, and block rewards at current FIL prices are unlikely to cover operating expenses for many SPs. In this environment, extending a flat historical renewal rate produces trajectories that contradict on-chain reality. The declining renewal rate curves used here reflect the expectation that SPs will increasingly choose not to renew as the economics deteriorate.
- **Known model bias (optimistic):** The model's renewal mechanism recalculates pledge at current conditions and takes the maximum of original and recalculated pledge (matching protocol behavior). This creates a mechanical effect where renewed sectors lock *more* FIL than they originally did. For example, with 374k FIL/day in scheduled pledge expirations and a 73% renewal rate: the 27% that don't renew release ~101k FIL/day, but the 73% that do renew have their pledge recalculated upward by ~37% (reflecting the increase in pledge per sector over the past year), adding ~101k FIL/day in uplift — nearly offsetting the entire expiration loss. New onboarding then pushes the net flow positive. In practice, the rising pledge is itself what deters renewals — SPs who face substantially higher collateral requirements may simply choose to exit rather than pay. The model cannot capture this feedback: it applies the renewal rate *independently* of the pledge increase. The declining renewal rate curves used in our scenarios are our way of approximating this behavioral response exogenously. Even so, the on-chain locked FIL decline rate (-145k to -170k FIL/day) remains steeper than the model's Decline scenario first-90-day slope (-62k FIL/day). These projections should therefore be interpreted as an optimistic bound.